

Communication-Aware Adaptive Weights for Consensus-Oriented Distributed KLA-Based LMB Fusion

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ABSTRACT

Distributed multi-sensor multi-object tracking over peer-to-peer networks requires fusion rules that remain conservative under unknown cross-correlations while responding to geometry-induced sensing variation and heterogeneous communication loss. Kullback-Leibler average (KLA), or geometric-average fusion, provides a natural foundation, but fixed or topology-derived weights cannot express posterior informativeness or realized packet delivery. This paper proposes communication-aware adaptive weighting for consensus-oriented distributed KLA-based LMB fusion. The retained configuration combines covariance quality, realized link quality, and existence confidence, then applies a branch-aware refinement that decouples spatial and existence weighting with weak structure-aware modulation. A controlled 50-trial component study evaluates network disagreement, local tracking accuracy, and filter/fusion runtime. Realized link quality supplies the dominant gain, while the combined branch-aware refinement provides a smaller but repeatable spatial-consensus improvement. The Balanced mode reduces OSPA consensus error, matched localization disagreement, and cardinality dispersion by 28.8%, 35.5%, and 85.4% relative to Fixed Metropolis, while requiring about 1.10× the fixed runtime. A Cardinality-critical mode adds FID-FIA only to the existence branch and reduces cardinality dispersion by a further 34.8% relative to Balanced, at the cost of higher localization disagreement and local RMSE.

1. Introduction

Distributed multi-sensor multi-object tracking is a core capability in surveillance, autonomous systems, and cooperative perception. The difficult case considered in this paper is not centralized multi-view fusion with a reliable communication backbone, but peer-to-peer tracking in which each node has partial spatial coverage, local posterior quality varies over time, and communication is affected by bandwidth limits, packet drops, or temporary outages [1]. Random finite set (RFS), finite-set statistics (FISST), and labeled-RFS filters provide the Bayesian state representation needed for this setting because they model target number, kinematic state, and identity in a single multi-object posterior [2–6]. The fusion problem then becomes a second-stage question: after each node has formed a local labeled posterior, how should neighboring posteriors be combined when field-of-view geometry, posterior uncertainty, and packet delivery all vary across the peer-to-peer network?

Within the labeled-RFS family, GLMB and LMB filters offer a practical balance between multi-object modeling fidelity and computational tractability [5–9]. For distributed fusion, Kullback-Leibler average (KLA), equivalently geometric-average or generalized covariance-intersection fusion, is attractive because it gives a conservative logarithmic-opinion-pool rule when unknown cross-correlations make exact Bayesian fusion unsafe [10–15]. This makes KLA-based LMB fusion a natural backbone for peer-to-peer multi-object tracking. The weakness is not the conservative fusion principle itself, but the usual assumption that the fusion weights can be fixed, uniform, or derived primarily from network topology.

Choosing these weights is therefore the central practical bottleneck. Fixed Metropolis, uniform, or degree-based consensus weights [16] are easy to deploy, but they cannot distinguish a precise local posterior from an ambiguous one, and they cannot tell whether a nominally useful neighbor has actually delivered useful information at the current time step. Existing adaptive-weight ideas address parts of this issue. Centralized and complementary LMB fusion

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studies adapt sensor contributions under heterogeneous fields of view [17–19]; distributed RFS fusion has also begun to assign weights to local subdensity components in peer-to-peer networks [20]. Information-geometric and Fisher-information weighting further show that local information content can be a principled weighting signal [21, 22]. Existing adaptive weighting rules are often organized around a dominant cue, such as graph topology, local information content, detectability, or Fisher-information-based target separability. The missing piece is not another isolated cue, but a factorized and branch-aware weighting framework for communication-constrained peer-to-peer GA-LMB fusion that can systematically evaluate and combine complementary reliability cues rather than forcing a single scalar signal to govern the whole posterior.

This general gap becomes especially visible in the two-part structure of the LMB posterior: a single scalar reliability score is unlikely to characterize both the spatial accuracy of a Bernoulli component and the decisiveness of its existence probability. In GA-LMB fusion, a node may provide a well-localized state estimate while remaining uncertain about whether the corresponding object exists, or it may provide decisive existence evidence whose spatial density is still comparatively diffuse.

Motivated by this gap, this paper proposes communication-aware adaptive fusion-weight allocation for distributed GA-LMB fusion. The design builds a shared quality backbone from covariance quality, realized link quality, and existence confidence, then separates the spatial and existence branches of the LMB fusion rule. A weak structure-aware correction regularizes spatial fusion without letting topology dominate the posterior-quality terms. This produces the Balanced mode. A Cardinality-critical mode then adds an FID-FIA-informed cue based on Cao and Zhao’s Fisher-information-accumulation idea [21] only to the existence branch. Figure 1 summarizes the peer-to-peer update, communication, and branch-decoupled fusion workflow.

The evaluation follows the same logic. The main experiment is an eight-sensor dual-formation tracking scenario under tiered heterogeneous packet loss, chosen to stress geometry/FOV-induced posterior-quality variation and realized communication heterogeneity. Network-level disagreement metrics are the primary outcome because the method is designed to reduce disagreement in the fused multi-sensor picture; truth-referenced local metrics and filter/fusion runtime are reported as safeguards against hidden accuracy or deployment-cost regressions. The controlled factor study shows that realized link quality explains the dominant gain, covariance provides an additional cardinality benefit, and the combined branch-aware refinement supplies a smaller but repeatable spatial-consensus improvement. The existence-only FID-FIA extension then exposes a clear operating tradeoff: lower cardinality dispersion and local cardinality error, but higher localization disagreement and local RMSE.

The main contributions of this paper are as follows.

1. We propose a communication-aware adaptive fusion-weight allocation family for distributed KLA-based GA-LMB fusion under communication constraints.
2. We introduce a factorized quality model that combines posterior covariance, realized link quality, and existence confidence, together with branch-decoupled weak structure-aware modulation.
3. We provide a 50-trial paired component study that separates single-factor effects from the progressive covariance-link, existence-confidence, and branch-aware design path instead of attributing the full fixed-baseline gain to every internal mechanism.
4. We show that the Balanced configuration improves network-level consensus over fixed and partial-factor baselines with about 1.10× fixed filtering/fusion runtime, while the Cardinality-critical extension supplies a further target-number gain with an explicit spatial-accuracy and computational-cost tradeoff.

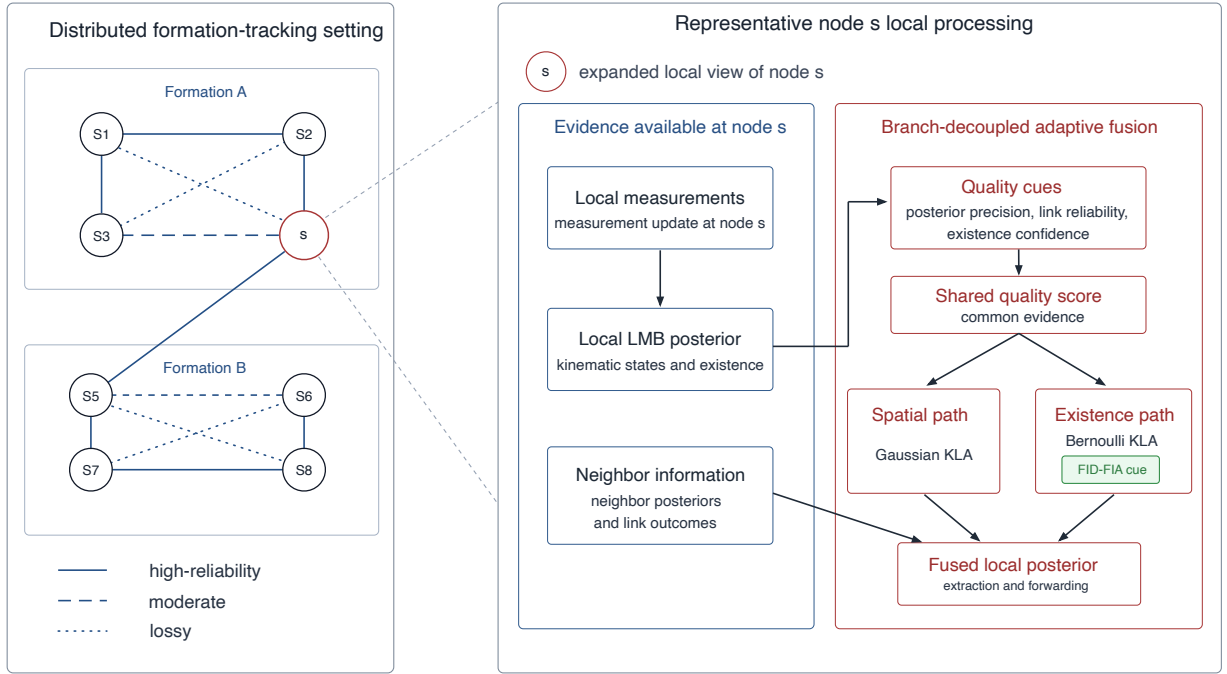


Figure 1: System overview of the distributed eight-sensor tracking setting, including local measurement update, neighborhood communication, branch-decoupled adaptive GA-LMB fusion, and the Cardinality-critical FID-FIA extension on the existence-weight path.

2. Related Work

The most relevant prior work falls into three groups: labeled-RFS filtering, conservative distributed density fusion, and adaptive or weighted RFS/LMB fusion. The first group provides the statistical state representation. Mahler's RFS/FISST formulation and PHD recursion establish the set-valued Bayesian foundation [2–4], while GLMB and LMB filters add labeled multi-object posteriors suitable for track identity and tractable implementation [5–9]. The present paper does not modify this filtering family; it uses LMB posteriors as the objects to be fused.

The second group concerns conservative fusion when cross-correlations are unknown. Covariance intersection, Kullback-Leibler averaging, geometric-average fusion, and inverse-covariance-intersection variants provide density-combination rules that avoid overconfident fusion under unknown dependence [10–13, 23–25]. Related logarithmic-opinion-pool and exponential-mixture formulations connect the same idea to distributed Bayesian fusion and consensus [14, 15]. In the labeled-RFS setting, GCI/KLA-style fusion has been extended to multi-Bernoulli and labeled-RFS densities, including work on label mismatch and peer-to-peer resilience [26–28]. These studies justify the GA/KLA backbone used here, but they leave open how the weights should respond to time-varying posterior quality and realized communication quality.

The third group is most directly related to adaptive weighting. Centralized multiple-view LMB fusion shows that information-aware weights can help when sensors have limited or different fields of view [17, 18]. Klupacs et al. take a complementary-fusion view, fusing label-associated Bernoulli densities and tuning sensor contribution by object detectability [19]. Consensus-based distributed LMB tracking has also explored automatic track-level weight evolution for different fields of view [29]. Recent peer-to-peer RFS work moves still closer to the present setting by factorizing local RFS densities into subdensities and assigning local weighting coefficients before GA or AA fusion under consensus or flooding protocols [20]. The closest information-geometric GA/KLA precedent is Cao and Zhao's distributed heterogeneous labeled-RFS method, which uses Fisher information distance and Fisher information accumulation (FID-FIA) to assign adaptive scalar fusion weights under heterogeneous sensing architectures [21]. In parallel, AA density fusion and related minimum-information-loss or cardinality-consistent formulations clarify the

theoretical contrast between AA and GA fusion [30–33]. A recent TechRxiv preprint in the AA-density-fusion series further studies Fisher-information-proportional weights for time-aligned asynchronous multi-rate filters and reports their use in both AA and GA fusion [22]. These works collectively support adaptive information-aware weighting, but they do not combine realized packet delivery, LMB branch decoupling, and existence-only information-geometric refinement in the synchronous peer-to-peer GA-LMB packet-loss setting studied here.

The present method builds on these precedents but narrows the design question. Rather than switching to complementary union, replacing the whole GA-LMB posterior by one FID-FIA scalar weight, or solving asynchronous time alignment, it retains the distributed KLA-based GA-LMB operator and changes the weight-allocation mechanism around it. The Balanced configuration uses posterior concentration, realized packet delivery, existence decisiveness, branch decoupling, and weak structure awareness. The Cardinality-critical mode confines the information-geometric cue to existence fusion and evaluates it separately, so its cardinality effect is not conflated with the lower-cost communication-aware backbone.

Communication constraints provide the deployment motivation for the weighting problem. Bandwidth limits, delays, asynchronous updates, and packet losses are central design factors in distributed fusion over sensor networks [1]. The present work focuses on the packet-delivery side of this problem: realized communication quality enters directly into the fusion weights, rather than appearing only as a nominal graph topology or an external simulation condition.

Against this background, the contribution of this paper is deliberately narrower than a broad new fusion framework. We stay within distributed KLA-based GA-LMB fusion, replace fixed or topology-only weighting with a covariance, realized-link, and existence-confidence backbone, and add branch-specific refinements for spatial and existence fusion. This positioning differentiates the method from both fixed-weight KLA fusion and adaptive scalar-weight formulations that do not explicitly decouple the LMB branches under heterogeneous packet loss.

3. Problem Formulation

Consider a distributed sensor network with sensor index set $\mathcal{V} = \{1, \dots, S\}$ operating over discrete time $k \in \{1, \dots, T\}$. The multi-object state at time k is modeled as a labeled random finite set X_k , and each sensor $s \in \mathcal{V}$ acquires a local measurement set $Z_k^{(s)}$. Following the standard RFS/FISST and labeled-RFS formulation, each node maintains an LMB approximation to the local multi-object posterior, whose Bernoulli components jointly encode target existence, kinematic state, and identity [2, 4–7]. The objective of the distributed fusion layer is to combine neighboring local posteriors into a consensus-quality posterior without assuming that cross-correlations among nodes are known.

3.1. Local LMB posterior model

Let $\mathcal{N}_s \subseteq \mathcal{V}$ denote the communication neighborhood of sensor s , including the node itself. In the setting considered here, $\mathcal{N}_s$ is induced by a communication graph derived from sensor geometry and communication range, and each node performs local fusion only over the posteriors available inside this neighborhood. After local prediction, measurement update, and data association, node $j \in \mathcal{N}_s$ produces a measurement-updated LMB posterior

$$\pi_k^{(j)} = \left\{ \left(r_{k,i}^{(j)}, p_{k,i}^{(j)}(\cdot, \ell_i) \right) \right\}_{i=1}^{M_k}, \quad (1)$$

where $r_{k,i}^{(j)} \in [0, 1]$ is the Bernoulli existence probability and $p_{k,i}^{(j)}$ is the associated spatial density for label ℓ_i .

3.2. Communication-constrained delivery

Before local measurements are used by a neighboring fusion node, they pass through a communication layer with three possible effects:

1. a global or per-step bandwidth budget that limits how many measurements can be delivered,
2. a stochastic packet-drop model on each sensor link,
3. optional node-level outages.

Accordingly, the posterior that reaches a neighbor is influenced not only by the underlying sensing quality, but also by whether measurements were actually delivered. Let $m_k^{(j)} \in \{0, 1\}$ denote the availability indicator of node j at time k , and let $\rho_k^{(j)} \in [0, 1]$ denote the realized link-quality statistic derived from delivered-versus-dropped packets.

3.3. Distributed KLA fusion objective

Because the cross-correlation structure among neighboring LMB posteriors is unknown, exact Bayesian fusion is generally unavailable. We therefore adopt a conservative Kullback-Leibler-average or geometric-average fusion rule. At node s , the target fused density over the neighborhood $\mathcal{N}_s$ is

$$\bar{\pi}_k^{(s)}(X) \propto \prod_{j \in \mathcal{N}_s} \left(\pi_k^{(j)}(X) \right)^{\omega_{k,s}^{(j)}}, \quad \sum_{j \in \mathcal{N}_s} \omega_{k,s}^{(j)} = 1, \quad \omega_{k,s}^{(j)} \geq 0, \quad (2)$$

where $\omega_{k,s}^{(j)}$ is the fusion weight assigned by node s to neighbor j at time k [12, 13]. With fixed-weight fusion, $\omega_{k,s}^{(j)}$ is often chosen as a uniform or Metropolis-type consensus weight [16]. The central problem studied here is how to choose these weights adaptively so that they reflect both posterior quality and communication quality.

In the GA-LMB fusion rule considered here, the spatial and existence parts of each Bernoulli component are fused separately after moment matching. This separation makes it natural to ask whether the same scalar weight should govern both spatial fusion and existence fusion.

3.4. Adaptive weight-allocation objective

The adaptive weighting design starts from a shared quality backbone. For node s and neighbor $j \in \mathcal{N}_s$, define the unnormalized score

$$\tilde{\omega}_{k,s}^{(j)} = m_k^{(j)} \cdot q_{\text{cov},k}^{(j)} \cdot q_{\text{link},k}^{(j)} \cdot q_{\text{exist},k}^{(j)}, \quad (3)$$

where $m_k^{(j)}$ is the availability mask, $q_{\text{cov},k}^{(j)}$ is a covariance-quality score, $q_{\text{link},k}^{(j)}$ is a realized link-quality score, and $q_{\text{exist},k}^{(j)}$ is an existence-confidence score. The normalized fusion weight is then

$$\omega_{k,s}^{(j)} = \frac{\tilde{\omega}_{k,s}^{(j)}}{\sum_{u \in \mathcal{N}_s} \tilde{\omega}_{k,s}^{(u)}}. \quad (4)$$

3.5. Decoupled and weakly structure-aware weighting

The branch-decoupled fusion rule does not force the spatial and existence branches to share exactly the same dynamics. Instead, it forms branch-specific scores,

$$\tilde{\omega}_{k,s}^{x,(j)} = \text{blend}\left(\tilde{\omega}_{k,s}^{(j)}, \tilde{\omega}_{k,s,\text{sp}}^{(j)}, \eta_x\right), \quad \tilde{\omega}_{k,s}^{r,(j)} = \text{blend}\left(\tilde{\omega}_{k,s}^{(j)}, \tilde{\omega}_{k,s,\text{ex}}^{(j)}, \eta_r\right), \quad (5)$$

and then applies only a weak graph-aware correction,

$$\tilde{\omega}_{k,s}^{x,(j)} \leftarrow \tilde{\omega}_{k,s}^{x,(j)} \cdot \left(\xi_{k,s}^{x,(j)} \right)^{\gamma_x}, \quad \tilde{\omega}_{k,s}^{r,(j)} \leftarrow \tilde{\omega}_{k,s}^{r,(j)} \cdot \left(\xi_{k,s}^{r,(j)} \right)^{\gamma_r}. \quad (6)$$

Here $\xi_{k,s}^{x,(j)}$ and $\xi_{k,s}^{r,(j)}$ are local structure priors derived from neighborhood overlap and communication reliability. The existence-side exponent is chosen much smaller than the spatial-side exponent, $\gamma_r \ll \gamma_x$, so topology acts as a mild prior layered on top of posterior-quality and communication-quality signals rather than as the primary source of fusion weights.

3.6. Evaluation target

The aim of the above formulation is not only to improve each node's local tracking output, but also to improve agreement among nodes. This distinction is central in the present problem. Under unknown cross-correlations and communication losses, distributed fusion is useful only if neighboring nodes move toward a shared multi-sensor picture rather than drifting toward mutually inconsistent local beliefs.

Accordingly, the evaluation uses two complementary metric layers:

- primary network-level disagreement metrics, which quantify consensus error among the fused outputs of different sensors,

- secondary truth-referenced local tracking metrics, which verify that consensus-error reductions are not obtained by materially degrading conventional tracking accuracy.

Let $\hat{X}_k^{(s)}$ denote the extracted multi-object estimate at sensor s and time k , and let $\hat{n}_k^{(s)} = |\hat{X}_k^{(s)}|$ denote its estimated cardinality. The paper defines the following network-level disagreement metrics:

$$D_{\text{OSPA}}(k) = \frac{2}{S(S-1)} \sum_{1 \leq a < b \leq S} d_{\text{OSPA}}\left(\hat{X}_k^{(a)}, \hat{X}_k^{(b)}\right), \quad (7)$$

$$D_{\text{loc}}(k) = \frac{2}{S(S-1)} \sum_{1 \leq a < b \leq S} d_{\text{loc}}\left(\hat{X}_k^{(a)}, \hat{X}_k^{(b)}\right), \quad (8)$$

$$D_{\text{card}}(k) = \frac{1}{S} \sum_{s=1}^S \left| \hat{n}_k^{(s)} - \text{median}\left(\hat{n}_k^{(1)}, \dots, \hat{n}_k^{(S)}\right) \right|, \quad (9)$$

where d_{OSPA} is the pairwise OSPA distance between two estimated finite sets and d_{loc} is the matched localization disagreement obtained after Hungarian matching on the common extracted objects [34–36]. In the paper these are reported as OSPA consensus error, matched localization disagreement, and cardinality dispersion, respectively.

These three quantities are task-driven network-level disagreement measures rather than standard field-wide tracking benchmarks. Their building blocks are standard or conventional: OSPA and related GOSPA-style distances are mainstream finite-set tracking metrics, Hungarian matching is the standard assignment primitive behind many tracking scores, and pairwise dispersion is the usual way to quantify disagreement in consensus problems [16, 34, 35, 37]. What is specific to this paper is the aggregation target: instead of comparing each sensor only with ground truth, the metrics compare the post-fusion outputs of different nodes. They are introduced because the present problem is distributed fusion under heterogeneous communication, where the main question is whether neighboring nodes move toward a common fused picture. For that reason, the paper does not use them as replacements for conventional truth-referenced tracking metrics, and it pairs the network-level disagreement metrics with the secondary local metrics listed below.

The secondary local metrics remain the standard truth-referenced tracking quantities:

- local E-OSPA,
- local RMSE,
- local cardinality error.

The evaluation logic is therefore asymmetric by design: the method is primarily successful if it reduces network-level disagreement, while the local metrics are used as a safeguard to show that this gain is not purchased by a clear loss in conventional tracking quality.

4. Communication-Aware Adaptive KLA Fusion

This section describes the adaptive fusion rule used in the distributed GA-LMB filter. The starting point is a conservative Kullback-Leibler-average or geometric-average fusion strategy for neighboring LMB posteriors under unknown cross-correlations [12, 13]. The main methodological question is not how to redesign the underlying labeled-RFS filter family, but how to allocate fusion weights so that they reflect posterior quality and realized communication quality in a communication-constrained sensor network. A variational and information-geometric interpretation of the branch-decoupled update is given in Appendix C.

4.1. Baseline GA-LMB fusion

At node s , let $\mathcal{N}_s$ denote the local communication neighborhood and let $\pi_k^{(j)}$ be the measurement-updated LMB posterior provided by node $j \in \mathcal{N}_s$. The baseline distributed fusion target is the weighted geometric average

$$\bar{\pi}_k^{(s)}(X) \propto \prod_{j \in \mathcal{N}_s} \left(\pi_k^{(j)}(X) \right)^{\omega_{k,s}^{(j)}}, \quad \sum_{j \in \mathcal{N}_s} \omega_{k,s}^{(j)} = 1, \quad \omega_{k,s}^{(j)} \geq 0. \quad (10)$$

After moment projection, the Gaussian approximation of the i th Bernoulli spatial density at node j is

$$p_{k,i}^{(j)}(x) \approx \mathcal{N}\left(x; v_{k,i}^{(j)}, T_{k,i}^{(j)}\right). \quad (11)$$

Using spatial fusion weights $\omega_{k,s}^{x,(j)}$, the fused spatial density is obtained in canonical form:

$$K_{k,i}^{(s)} = \sum_{j \in \mathcal{N}_s} \omega_{k,s}^{x,(j)} \left(T_{k,i}^{(j)}\right)^{-1}, \quad h_{k,i}^{(s)} = \sum_{j \in \mathcal{N}_s} \omega_{k,s}^{x,(j)} \left(T_{k,i}^{(j)}\right)^{-1} v_{k,i}^{(j)}, \quad (12)$$

$$\Sigma_{k,i}^{(s)} = \left(K_{k,i}^{(s)}\right)^{-1}, \quad \mu_{k,i}^{(s)} = \Sigma_{k,i}^{(s)} h_{k,i}^{(s)}. \quad (13)$$

Using existence fusion weights $\omega_{k,s}^{r,(j)}$, the Bernoulli existence probability is fused by the corresponding weighted geometric rule:

$$r_{k,i}^{(s)} = \frac{\eta_{k,i}^{(s)} \prod_{j \in \mathcal{N}_s} \left(r_{k,i}^{(j)}\right)^{\omega_{k,s}^{r,(j)}}}{\eta_{k,i}^{(s)} \prod_{j \in \mathcal{N}_s} \left(r_{k,i}^{(j)}\right)^{\omega_{k,s}^{r,(j)}} + \prod_{j \in \mathcal{N}_s} \left(1 - r_{k,i}^{(j)}\right)^{\omega_{k,s}^{r,(j)}}}, \quad (14)$$

where $\eta_{k,i}^{(s)}$ is the Gaussian normalization term induced by the fused spatial density.

4.2. Factorized adaptive weight backbone

The adaptive design begins with a shared quality backbone. For node s and neighbor $j \in \mathcal{N}_s$, define the raw score

$$\tilde{\omega}_{k,s}^{(j)} = m_k^{(j)} \cdot q_{\text{cov},k}^{(j)} \cdot q_{\text{link},k}^{(j)} \cdot q_{\text{exist},k}^{(j)}, \quad (15)$$

where $m_k^{(j)} \in \{0, 1\}$ is the availability mask, $q_{\text{cov},k}^{(j)}$ is a covariance-quality score, $q_{\text{link},k}^{(j)}$ is a realized link-quality score, and $q_{\text{exist},k}^{(j)}$ is an existence-confidence score. The normalized weight is

$$\bar{\omega}_{k,s}^{(j)} = \frac{\tilde{\omega}_{k,s}^{(j)}}{\sum_{u \in \mathcal{N}_s} \tilde{\omega}_{k,s}^{(u)}}. \quad (16)$$

This factorization focuses on the three factors that are most directly aligned with the design objective: posterior concentration through covariance, communication reliability through delivered-versus-dropped packets, and existence decisiveness through Bernoulli existence probabilities. Figure 2 shows how this shared backbone feeds the decoupled spatial and existence-weight paths.

4.3. Covariance and link-quality terms

For node j , let $T_{k,i}^{(j)}$ be the moment-matched covariance of the i th Bernoulli component. The covariance-quality score is defined as

$$q_{\text{cov},k}^{(j)} = \frac{1}{\epsilon + \frac{1}{M_k} \sum_{i=1}^{M_k} \text{tr}\left(T_{k,i}^{(j)}\right)}, \quad (17)$$

so that a node with smaller posterior spread receives a larger score.

The link-quality term is derived from realized communication outcomes rather than nominal topology. Let $d_k^{(j)}$ and $\ell_k^{(j)}$ denote the counts of delivered and dropped measurements, respectively. The realized link-quality score is defined as

$$q_{\text{link},k}^{(j)} = \frac{d_k^{(j)}}{d_k^{(j)} + \ell_k^{(j)}}, \quad (18)$$

which directly reflects how reliably that node's information is reaching its neighbors.

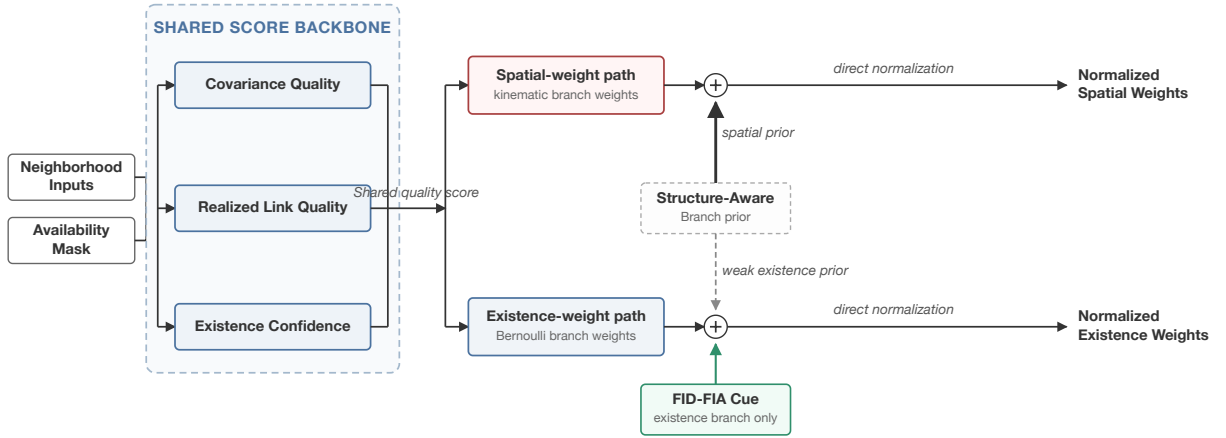


Figure 2: Adaptive fusion-weight factorization, showing the shared quality backbone, decoupled spatial and existence-weight paths, weak structure-aware branch prior, and an FID-FIA cue applied only to the existence branch.

4.4. Existence-confidence weighting

To capture the missing existence-related dimension, the method computes a certainty score from Bernoulli existence probabilities. For the i th Bernoulli component at node j ,

$$c_{k,i}^{(j)} = \left| 2r_{k,i}^{(j)} - 1 \right|. \quad (19)$$

The per-node weighted existence confidence is then

$$\bar{c}_k^{(j)} = \frac{\sum_{i=1}^{M_k} r_{k,i}^{(j)} c_{k,i}^{(j)}}{\epsilon + \sum_{i=1}^{M_k} r_{k,i}^{(j)}}. \quad (20)$$

The final existence-confidence score is mapped into a bounded positive factor:

$$q_{\text{exist},k}^{(j)} = \lambda_{\min} + (1 - \lambda_{\min}) \left(\bar{c}_k^{(j)} \right)^{p_e}, \quad (21)$$

where $\lambda_{\min} \in (0, 1]$ and $p_e > 0$ are tunable parameters. Figure 3 visualizes this bounded Bernoulli-level mapping and representative node-level aggregation effects.

4.5. Decoupled spatial and existence scores

The proposed method does not force the same adaptive score to govern spatial fusion and existence fusion. This separation is structurally motivated by the Bernoulli/LMB form itself: existence and spatial terms enter the fused posterior through different update blocks, even though the formal variational interpretation is deferred to Appendix C. The method therefore constructs branch-specific dedicated scores:

$$\tilde{w}_{k,s,\text{sp}}^{(j)} = m_k^{(j)} \cdot \left(q_{\text{cov},k}^{(j)} \right)^{\alpha_x} \cdot \left(q_{\text{link},k}^{(j)} \right)^{\beta_x}, \quad (22)$$

$$\tilde{w}_{k,s,\text{ex}}^{(j)} = m_k^{(j)} \cdot \left(q_{\text{link},k}^{(j)} \right)^{\beta_r} \cdot \left(q_{\text{exist},k}^{(j)} \right)^{\alpha_r}. \quad (23)$$

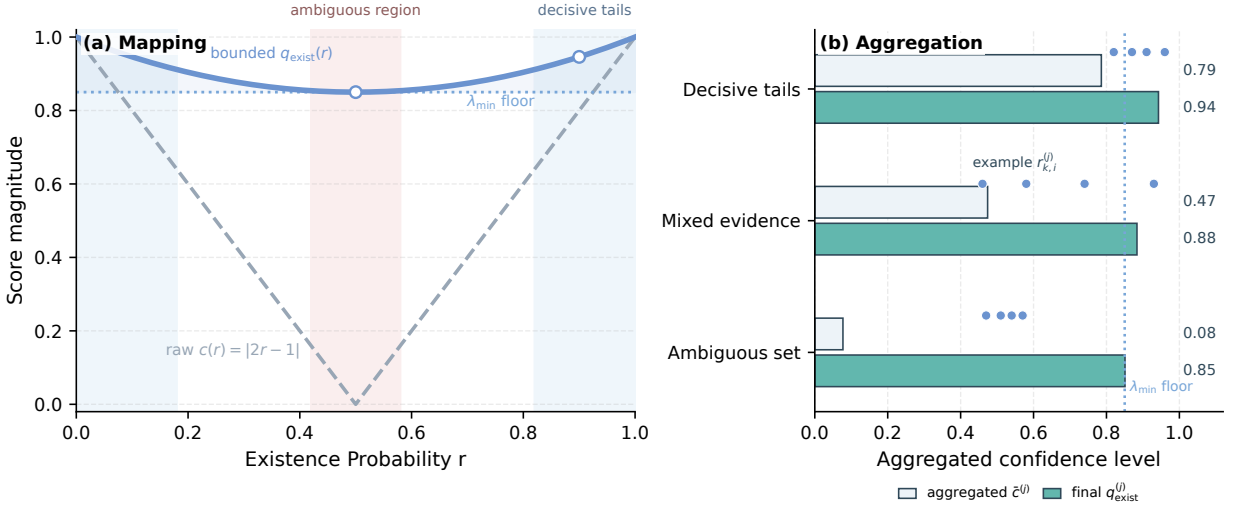


Figure 3: Existence-confidence weighting used in the adaptive rule: (a) the Bernoulli-level decisiveness mapping and bounded score curve, and (b) representative node-level aggregation examples showing how decisive posterior sets produce larger $\tilde{c}^{(j)}$ and $q_{\text{exist}}^{(j)}$.

The final branch scores are obtained through geometric interpolation:

$$\tilde{\omega}_{k,s}^{x,(j)} = \left(\tilde{\omega}_{k,s}^{(j)}\right)^{1-\eta_x} \left(\tilde{\omega}_{k,s,\text{sp}}^{(j)}\right)^{\eta_x}, \quad \tilde{\omega}_{k,s}^{r,(j)} = \left(\tilde{\omega}_{k,s}^{(j)}\right)^{1-\eta_r} \left(\tilde{\omega}_{k,s,\text{ex}}^{(j)}\right)^{\eta_r}. \quad (24)$$

The point of this decoupling is not to create two unrelated methods, but to acknowledge that spatial consensus and existence consensus respond differently to weight perturbations. It also determines how Fisher-information-based cues are used in this paper. The information-geometric construction of Cao and Zhao is a natural scalar indicator of local target-separability and posterior informativeness, and in our experiments it is especially effective for cardinality and existence decisions [21]. Related multi-rate average-fusion work also supports Fisher information as a principled local information measure after time alignment [22]. However, applying the same Fisher-type scalar weight to the whole fused posterior can trade away spatial RMSE in the present GA-LMB setting. In the log-utility view developed in Appendix C, a single FID-FIA scalar-weight baseline is a constrained case that forces the same Fisher-separability utility into both the Gaussian spatial barycenter and the Bernoulli existence pool. The branch-decoupled formulation relaxes this constraint: the communication-aware covariance/link-quality branch continues to govern spatial fusion, while the FID-FIA signal is reserved for the existence branch where its empirical strength is most aligned with the fused Bernoulli variable.

4.6. FID-FIA existence refinement

The Cardinality-critical information-geometric extension acts only on the existence branch. It follows the FID-FIA intuition of accumulating pairwise Fisher-information distance between locally represented targets, but it is not used as a replacement scalar fusion rule. For each node j , the method first computes a target-pair accumulation score $a_k^{(j)}$ from the local measurement-updated Bernoulli components. Each Bernoulli component is moment-projected, target pairs are weighted by their existence probabilities, and the pairwise distance is accumulated along the line segment between the two projected states using the local Fisher metric induced by the sensor's detection probability and measurement covariance. The score is then normalized over the available local neighborhood,

$$\hat{a}_{k,s}^{(j)} = \frac{a_k^{(j)}}{\epsilon + \max_{u \in \mathcal{N}_s} a_k^{(u)}}, \quad 0 \leq \hat{a}_{k,s}^{(j)} \leq 1. \quad (25)$$

This normalized accumulation is mapped into a bounded existence-only modulation factor,

$$q_{\text{fid},k,s}^{(j)} = \rho_{\min} + (1 - \rho_{\min}) \left(\hat{a}_{k,s}^{(j)} \right)^{p_f}. \quad (26)$$

Only the existence score is modified:

$$\tilde{\omega}_{k,s}^{r,(j)} \leftarrow \tilde{\omega}_{k,s}^{r,(j)} \left(q_{\text{fid},k,s}^{(j)} \right)^{\gamma_f}. \quad (27)$$

The spatial score $\tilde{\omega}_{k,s}^{x,(j)}$ is left unchanged. This update adds a strong existence-only modulation without replacing the communication-aware spatial score. Equivalently, for positive $q_{\text{fid},k,s}^{(j)}$, the update adds $\gamma_f \log q_{\text{fid},k,s}^{(j)}$ to the existence-branch utility and adds nothing to the spatial utility.

4.7. Retained operating configuration

The retained *Balanced mode* uses the three-factor adaptive backbone, decoupled spatial and existence scores, and weak structure-aware refinement.

The *Cardinality-critical mode* inherits the Balanced configuration and changes only the existence score through the FID-FIA cue. The 50-trial evaluation shows that this mode further reduces cardinality dispersion and local cardinality error, while increasing localization disagreement, local RMSE, and computation relative to Balanced. The two modes are therefore retained for different deployment priorities rather than ranked by a single aggregate score.

4.8. Weak structure-aware refinement

After branch decoupling and before the final FID-FIA existence modulation, the method applies a weak graph-aware correction through local structure priors derived from neighborhood-overlap similarity and communication-reliability information. The existence-side structure strength is intentionally kept much smaller than the spatial-side structure strength, so topology provides only a secondary regularization signal and does not dominate the core weight assignment.

After score normalization, the resulting branch weights preserve responsiveness to realized packet delivery and allow a sufficiently weak neighbor to receive a near-zero weight.

4.9. Optional extension modules

The evaluated software also supports several exploratory extension modules outside the claimed core method. Their empirical effect is weaker, more coupled, or less stable than the evidence for the main three-factor backbone plus the retained branch-specific refinements, so they are kept out of the main method description and summarized only in Appendix B.

5. Experimental Setup

This section defines the evaluation protocol used to support the paper's claims. The primary evidence is the dual-formation eight-sensor distributed GA-LMB tracking problem under tiered heterogeneous packet loss. A controlled paired experiment isolates each retained component under one consistent configuration.

5.1. Evaluation hierarchy

The evaluation is intentionally hierarchical. The tiered-loss dual-formation scenario provides the confirmatory evidence for the proposed GA-LMB method family, while the factor ablation in the same scenario identifies which parts of the adaptive-weight design carry the result. Computational cost is measured on this main scenario as a deployment constraint rather than as a post hoc timing note. Two smaller supporting studies then test the boundaries of the claim: an ideal-communication comparison asks whether the refinement is useful without packet loss, and a communication-level sensitivity probe asks whether the same behavior persists as communication constraints become more severe. Additional exploratory variants are treated as appendix material rather than as competing headline methods.

5.2. Main scenario

The primary evaluation setting uses eight mobile sensors arranged as two four-sensor formations. The filtering mode is distributed local fusion, meaning that each node fuses only its own posterior and the posteriors available inside its communication neighborhood. The local multi-object filter is GA-LMB with loopy belief propagation (LBP) for approximate marginal data association [38].

The choice of the GA/KLA route is deliberate. GA fusion gives a conservative logarithmic-opinion-pool update under unknown cross-correlations, admits a stable canonical-form Gaussian fusion step for the LMB spatial densities, and naturally exposes the Bernoulli existence product rule in which branch-specific weighting can be applied. These properties make it a cleaner vehicle for studying adaptive weight allocation than a broad comparison across fusion families.

The common sensor-side configuration is:

- number of sensors: 8,
- filter mode: GA-LMB,
- data association: LBP,
- clutter rate: 3 for every sensor,
- detection probability: 0.9 for every sensor,
- measurement-noise scale: $q = 3$ for every sensor,
- sensor field of view enabled with half-angle 60° and range 60000.

The nominal detection probability, measurement-noise scale, and clutter rate are therefore homogeneous across sensors. The sensing variation exercised by the main scenario comes from the dual-formation geometry, finite fields of view, target visibility, and the resulting posterior covariance and existence differences; it should not be read as a sensitivity test over explicit p_D , measurement-noise, or clutter tiers.

The target side consists of ten objects arranged in three groups with staggered births. The three groups contain $3 + 3 + 4$ targets, respectively, and the targets move toward the central region with nominal speed 0.45. Births are staggered every eight time steps, and the total simulation length is 100 steps.

5.3. Communication model

The main scenario uses a tiered heterogeneous packet-loss setting. Communication is applied before distributed fusion through a measurement-delivery model with bandwidth limitation and link-level packet loss. The main configuration is:

- communication level: 2,
- global maximum delivered measurements per step: 80,
- packet-selection policy: weighted priority with random measurement selection,
- link model: fixed packet drop,
- target mean drop rate: $p_{\text{drop}} = 0.2$,
- tiered drop levels: $[0, 0.1, 0.2, 0.5]$,
- tiered level counts: $[1, 4, 1, 2]$.

This tiered design preserves the historical mean communication degradation while introducing persistent cross-node heterogeneity.

Figure 4 summarizes the contrast between the simpler standard fixed ideal benchmark and the main tiered-loss scenario. The standard ideal benchmark uses four full-ROI sensors, ideal communication, and fixed target births. The main scenario is the dual-formation eight-sensor case with staggered target waves, a bandwidth cap, and persistent drop-tier heterogeneity before distributed fusion.

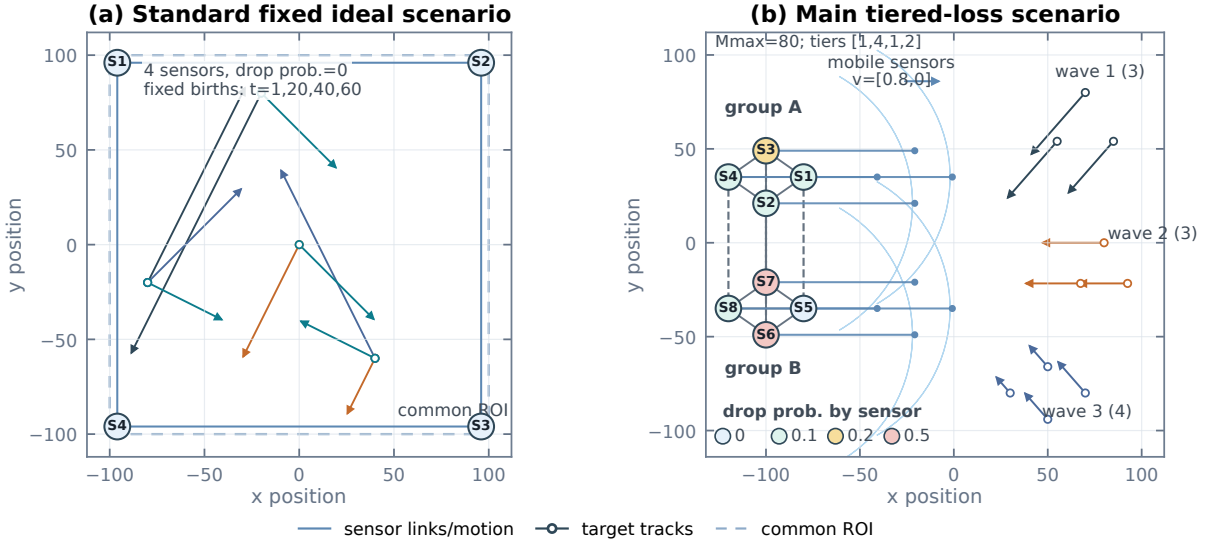


Figure 4: Schematic comparison between the standard fixed ideal benchmark and the main tiered heterogeneous packet-loss scenario. The ideal benchmark uses four full-ROI sensors with no packet loss and fixed target births; the main scenario uses the dual-formation eight-sensor geometry and staggered target waves with the bandwidth cap and persistent drop-tier heterogeneity used in the main experiments.

5.4. Mainline ablation protocol

The core ablation study compares the following controlled variants:

1. Fixed Metropolis,
2. covariance only, link quality only, and existence confidence only,
3. covariance + link,
4. covariance + link + existence confidence with shared weights,
5. branch-aware spatial/existence refinement with weak structure-aware modulation, retained as the Balanced mode,
6. Cardinality-critical mode, which adds FID-FIA only to the existence branch of Balanced.

The Balanced mode uses existence-confidence parameters $\lambda_{\min} = 0.85$ and $p_e = 2.0$, decoupled KLA strengths (0.5, 0.15) for the spatial and existence branches, and weak structure-aware strengths (0.45, 0.08). The Cardinality-critical mode preserves this configuration and modifies only the existence branch.

5.5. Metrics and reporting

The experiments report three categories of metrics. The first category is network-level disagreement among sensors after distributed fusion. These primary metrics are OSPA consensus error, matched localization disagreement, and cardinality dispersion as defined in Section 3. The second category is conventional truth-referenced tracking accuracy. These secondary metrics are local E-OSPA, local RMSE, and local cardinality error. The third category is computational cost, measured as filter/fusion wall-clock runtime per trial, runtime per simulation step, and runtime relative to Fixed Metropolis. The OSPA/GOSPA family is used because it provides mathematically consistent finite-set distances for multi-object filtering and tracking evaluation [34, 35, 37]. The primary disagreement metrics are task-specific network-agreement measures built from these standard ingredients, not a replacement for standard truth-referenced tracking benchmarks, so the paper always interprets them together with the secondary local metrics and the computational-cost measurements rather than in isolation.

This hierarchy is deliberate. The proposed method is not intended as a generic single-sensor accuracy booster. It is designed as a distributed fusion-weight mechanism for reducing disagreement in the fused multi-sensor picture under

heterogeneous communication. For that reason, the paper treats the network-level disagreement metrics as the main outcome, the local tracking metrics as a safeguard against hidden degradation, and computational cost as a first-class deployment-feasibility axis. The target is to improve cross-node agreement without causing a meaningful drop in conventional tracking quality or requiring impractical additional computation. Runtime is measured only around the distributed LMB filtering/fusion call for each arm; scenario generation, communication-model sampling, plotting, and metric evaluation are excluded so that the reported cost isolates the algorithmic overhead of adaptive weighting and branch-specific FID-FIA scoring.

Unless otherwise stated, reported scalar metrics are Monte Carlo means from 50 deterministic paired trials with seeds 2–51. Every arm in the main component experiment shares the same truth, measurements, and communication realization within each trial.

6. Results and Discussion

This section separates two kinds of evidence. Tables 1–3 compare complete operating points against fixed and direct adaptive baselines. Table 4 is the controlled component study used for factor attribution. All reported accuracy studies use 50 deterministic trials with seeds 2–51 in the same tiered heterogeneous packet-loss scenario. The direct PD- and FID-FIA-weighted baselines come from dedicated matched-seed probes, so incremental paired claims are reserved for Table 4.

6.1. Main result in the tiered-drop GA scenario

Table 1 is the main method-level comparison. PD-weighted GA tests whether a direct sensor-reliability weight is sufficient, while FID-FIA-weighted GA applies one information-geometric weight to the complete posterior. The two proposed rows instead use branch-aware communication-adaptive fusion. Balanced gives the lowest OSPA consensus error and matched localization disagreement. Cardinality-critical adds FID-FIA only to the existence branch and gives the lowest cardinality dispersion.

Relative to Fixed Metropolis, the Balanced mode reduces OSPA consensus error by 28.8%, matched localization disagreement by 35.5%, and cardinality dispersion by 85.4%. It also improves the two spatial disagreement metrics over both direct baselines. Cardinality-critical reduces cardinality dispersion by a further 34.8% relative to Balanced, but its OSPA consensus error increases by 1.0% and localization disagreement by 8.9%. The method-level table therefore exposes a spatial-versus-cardinality operating tradeoff rather than a uniformly dominant setting.

Table 2 checks whether stronger agreement is obtained by driving all nodes toward an inaccurate common estimate. PD-weighted GA has the lowest local RMSE, but remains weak on finite-set and cardinality errors. Balanced provides the stronger spatial-consensus operating point. Cardinality-critical improves local E-OSPA and reduces local cardinality error by 26.2% relative to Balanced, but increases local RMSE by 6.6%. This mirrors the network-level spatial-versus-cardinality tradeoff.

Computational cost is the third evaluation axis for this method family. Table 3 reports 50-trial runtime measurements in the same 100-step tiered-drop scenario and covers all arms in the main comparison. The measurement includes only the distributed LMB filtering/fusion call for each arm; scenario generation, communication-model sampling, plotting, and metric evaluation are excluded. The main cost distinction is clear. The Balanced mode remains close to the fixed-weight baseline, requiring about 1.08× the Fixed Metropolis runtime. The PD-weighted GA baseline is also a moderate-cost dynamic-weighting alternative at about 1.23× fixed runtime. By contrast, FID-FIA-weighted GA and the Cardinality-critical mode are both near the three-times-cost regime. The runtime evidence therefore identifies FID-FIA scoring, not the basic communication-aware weighting backbone or the direct PD-weighted baseline, as the dominant source of additional cost.

The cost results sharpen the operating-mode recommendation. The Balanced mode is the low-overhead default when spatial consensus, RMSE stability, runtime, or energy budget is the limiting requirement. The Cardinality-critical mode should be selected when target-number agreement and suppression of missed or false support justify roughly three times the fixed-weight filtering/fusion time.

6.2. Factor-by-factor ablation

Table 4 separates isolated factor effects from the progressive construction of the retained method. The component matrix makes each comparison explicit: the first block tests covariance, realized link quality, and existence confidence independently, while the second block starts from the covariance-link backbone, adds existence confidence with shared

Table 1

Main operating-point comparison under tiered heterogeneous packet loss. Values are mean $\pm$ standard deviation over 50 deterministic trials with seeds 2–51. PD-weighted GA and FID-FIA-weighted GA are direct adaptive-weighting baselines evaluated in dedicated probes. Lower values are better.

Arm	OSPA err.	Loc. disag.	Card. disp.
Fixed Metropolis	2.383 ± 0.176	2.263 ± 0.340	0.650 ± 0.223
PD-weighted GA	2.177 ± 0.161	1.995 ± 0.233	0.588 ± 0.170
FID-FIA-weighted GA	1.818 ± 0.056	1.643 ± 0.109	0.123 ± 0.023
Balanced mode	1.696 ± 0.046	1.461 ± 0.053	0.095 ± 0.025
Cardinality-critical mode	1.713 ± 0.049	1.590 ± 0.167	0.062 ± 0.016

Table 2

Truth-referenced local tracking metrics for the same 50 deterministic seeds as Table 1. Each trial is averaged over sensors before the across-trial mean and standard deviation are computed.

Arm	E-OSPA	RMSE	CardErr
Fixed Metropolis	2.781 ± 0.129	1.630 ± 0.049	1.364 ± 0.260
PD-weighted GA	2.736 ± 0.116	1.563 ± 0.061	1.255 ± 0.205
FID-FIA-weighted GA	2.185 ± 0.050	1.734 ± 0.094	0.388 ± 0.047
Balanced mode	2.072 ± 0.051	1.636 ± 0.045	0.283 ± 0.046
Cardinality-critical mode	2.030 ± 0.042	1.744 ± 0.161	0.209 ± 0.026

Table 3

Computational cost in the main tiered heterogeneous packet-loss scenario. Wall-clock time measures only the distributed LMB filtering/fusion call for each arm over 50 deterministic paired trials with seeds 2–51. The Fixed Metropolis, FID-FIA-weighted GA, Balanced, and Cardinality-critical rows come from the operating-mode runtime probe; the PD-weighted GA row comes from the direct dynamic-weighting probe under the same scenario and seed set. Relative runtime is computed against the paired Fixed Metropolis denominator within the corresponding probe.

Arm	Runtime (s)	Runtime/step (s)	Relative runtime
Fixed Metropolis	52.123 ± 7.932	0.521	1.000×
PD-weighted GA	61.668 ± 5.376	0.617	1.233×
FID-FIA-weighted GA	147.674 ± 23.957	1.477	2.833×
Balanced mode	56.378 ± 9.626	0.564	1.082×
Cardinality-critical mode	155.913 ± 18.220	1.559	2.991×

weights, and then applies the branch-aware refinement used by Balanced mode. This final component comprises spatial/existence branch decoupling together with weak structure-aware modulation. The last block adds existence-only FID-FIA to obtain the Cardinality-critical mode.

The detailed results support five bounded conclusions. First, realized link quality is the dominant factor in this packet-loss scenario. Second, covariance provides a useful independent signal and gives a further cardinality benefit when combined with link quality. Third, existence confidence has little isolated incremental effect on the covariance-link backbone. Fourth, the combined branch-aware refinement supplies a small, repeatable spatial benefit. Fifth, existence-only FID-FIA provides a substantial additional cardinality benefit but does not improve the spatial disagreement metrics. The table intentionally does not separate branch decoupling from its weak structure-aware modulation.

6.3. Scope of the revised evidence

The Balanced and Cardinality-critical configurations are confirmed here in the main tiered heterogeneous packet-loss scenario. Earlier ideal-communication, communication-level, and arithmetic-average probes predate the current operating modes and are therefore not used as supporting evidence. Repeating those studies with the current configuration remains a targeted validation step.

Table 4

Component ablation under tiered heterogeneous packet loss. Values are mean $\pm$ standard deviation over 50 deterministic trials with matched seeds. Cov., Link, Exist., Branch, and FID denote covariance quality, realized link quality, existence confidence, the branch-aware refinement, and existence-only FID-FIA modulation, respectively. The Branch component jointly includes spatial/existence decoupling and weak structure-aware modulation.

Variant	Active components					Network disagreement		
	Cov.	Link	Exist.	Branch	FID	OSPA	Loc.	Card.
<i>Reference</i>								
Fixed Metropolis	–	–	–	–	–	2.383 $\pm$ 0.176	2.263 $\pm$ 0.340	0.650 $\pm$ 0.223
<i>Single-factor diagnostics</i>								
Covariance only	✓	–	–	–	–	1.983 $\pm$ 0.120	1.877 $\pm$ 0.240	0.372 $\pm$ 0.114
Link quality only	–	✓	–	–	–	1.712 $\pm$ 0.049	1.474 $\pm$ 0.053	0.102 $\pm$ 0.026
Existence confidence only	–	–	✓	–	–	2.057 $\pm$ 0.143	1.857 $\pm$ 0.228	0.447 $\pm$ 0.152
<i>Progressive retained design</i>								
Covariance + link	✓	✓	–	–	–	1.708 $\pm$ 0.046	1.483 $\pm$ 0.059	0.092 $\pm$ 0.025
+ existence (shared)	✓	✓	✓	–	–	1.710 $\pm$ 0.045	1.481 $\pm$ 0.055	0.093 $\pm$ 0.026
Balanced mode	✓	✓	✓	✓	–	1.696 $\pm$ 0.046	1.461 $\pm$ 0.053	0.095 $\pm$ 0.025
<i>Existence-branch extension</i>								
Cardinality-critical mode	✓	✓	✓	✓	✓	1.713 $\pm$ 0.049	1.590 $\pm$ 0.167	0.062 $\pm$ 0.016

7. Conclusion

This paper studies adaptive weight allocation for distributed KLA-based LMB fusion under heterogeneous packet loss. The revised evidence supports a narrower conclusion than a monolithic “all modules help” claim. Realized link quality is the dominant factor in the tested communication-constrained scenario. Covariance quality provides an independent posterior-concentration signal and gives a further cardinality benefit when combined with link quality. The combined branch-aware refinement then supplies a small but repeatable improvement in OSPA consensus error and matched localization disagreement.

Existence confidence has little isolated incremental effect on the covariance-link backbone in the present experiment. Branch decoupling and weak structure-aware modulation are therefore evaluated together as one branch-aware refinement rather than assigned separate empirical claims. The detailed ablation makes this distinction explicit and separates architectural motivation from measured effect size.

Relative to Fixed Metropolis, the Balanced configuration reduces OSPA consensus error, matched localization disagreement, and cardinality dispersion by 28.8%, 35.5%, and 85.4%, respectively, with approximately 1.10 $\times$ fixed filtering/fusion runtime. Its truth-referenced local E-OSPA and cardinality error remain substantially below the fixed baseline, while local RMSE stays close to the covariance-link backbone.

The Cardinality-critical mode retains the same branch-aware backbone and adds FID-FIA only to the existence branch. Relative to Balanced, it reduces cardinality dispersion by 34.8%, local E-OSPA by 2.0%, and local cardinality error by 26.2%. The tradeoff is explicit: OSPA consensus error increases by 1.0%, localization disagreement by 8.9%, and local RMSE by 6.6%. Balanced is therefore the spatial-consensus and lower-cost mode, whereas Cardinality-critical is appropriate when target-number errors dominate.

The current operating modes are confirmed in the main tiered-loss scenario. Repeating ideal-communication, communication-level, and arithmetic-average studies is the next validation step. Further work should also consider richer sensor-parameter heterogeneity, asynchronous or multi-rate communication, and target-wise information measures once reliable label association across neighboring LMB posteriors is available.

A. Simulation Setup

This appendix records reproducibility details for the reported experiments, including the scenario geometry, target initialization, communication model, adaptive configurations, supporting ideal-communication variant, and seed control.

A.1. Main scenario overview

The primary experiment uses an eight-sensor distributed formation-tracking scenario composed of two four-sensor formations. Each node runs a local GA-LMB filter with loopy belief propagation (LBP) for approximate marginal data association [38], and distributed fusion is performed only over the local communication neighborhood.

The common filter-side settings are:

- number of sensors: 8,
- local filter mode: GA-LMB,
- data association: LBP,
- simulation length: 100 time steps,
- sensor communication range: 150,
- fixed-weight baseline fusion rule: Metropolis-type consensus weighting [16],
- clutter rate at every sensor: 3,
- detection probability at every sensor: 0.9,
- measurement-noise scale at every sensor: $q = 3$,
- field of view enabled with half-angle 60° and range 60000.

These are homogeneous nominal sensor settings. Geometry and field-of-view gating create target- and time-dependent visibility differences, but no explicit per-sensor p_D , noise, or clutter tiers are used in the main reported experiment.

A.2. Sensor geometry and neighborhood graph

The eight sensors are divided into two groups of four. Both groups follow constant-velocity platform trajectories with zero process noise and move from left to right with nominal velocity $[0.8, 0]^T$. The two formation centers are $[-80, 35]^T$ and $[-80, -35]^T$, respectively. Each group uses a four-agent leader-follower pattern with one lead sensor and three offset followers, with spacing 20.

The local communication graph is fixed. Each formation is fully connected internally, and the two formations are connected by one-to-one cross links: $1 \leftrightarrow 5$, $2 \leftrightarrow 6$, $3 \leftrightarrow 7$, and $4 \leftrightarrow 8$. The corresponding neighborhood sets are therefore $\{1, 2, 3, 4, 5\}$, $\{1, 2, 3, 4, 6\}$, $\{1, 2, 3, 4, 7\}$, $\{1, 2, 3, 4, 8\}$, $\{1, 5, 6, 7, 8\}$, $\{2, 5, 6, 7, 8\}$, $\{3, 5, 6, 7, 8\}$, and $\{4, 5, 6, 7, 8\}$.

A.3. Target initialization

The target side contains ten objects divided into three groups with staggered births. The birth interval is eight time steps, the birth start time is one, and the target lifetime is 100 time steps. The three groups are: three targets in a Triangle formation centered at $[70, 80]^T$ with spacing 30, three targets in a Triangle formation centered at $[80, 0]^T$ with spacing 25, and four targets in a four-agent leader-follower pattern centered at $[70, -80]^T$ with spacing 20. All three groups move toward the central region with speed 0.45.

A.4. Communication model

Communication degradation is applied before distributed fusion through a model that combines a global per-step measurement budget with optional link-level packet loss and optional node outages. The communication levels are:

- level 0: no communication restriction,
- level 1: global measurement budget only,
- level 2: level 1 plus link-level packet loss,
- level 3: level 2 plus node outages.

The main paper scenario uses level 2 with global maximum delivered measurements per step 80, fixed link loss, nominal mean packet-drop rate $p_{\text{drop}} = 0.2$, and tiered heterogeneous packet-loss configuration

$$p_{\text{drop}} \in [0, 0.1, 0.2, 0.5], \quad \text{counts} = [1, 4, 1, 2]. \quad (28)$$

This tiered configuration preserves the historical mean packet-drop rate while creating persistent node-to-node communication heterogeneity.

A.5. Main adaptive configurations

The Balanced mode used in the main tiered-drop comparison is:

- `emaAlpha` = 0 and `minWeight` = 0,
- spatial/existence EMA coefficients and final-weight floors all set to zero,
- covariance, link-quality, and existence-confidence terms enabled,
- `existenceConfidenceMinScore` = 0.85,
- `existenceConfidencePower` = 2.0,
- decoupled KLA enabled,
- weak structure-aware refinement enabled,
- FID-FIA refinement disabled,
- posterior-structure-consistency, NIS, and history disabled in the main configuration,
- spatial decoupling and structure strengths (0.5, 0.45),
- existence decoupling and structure strengths (0.15, 0.08).

A.6. Monte Carlo seed control

The main component experiment uses 50 deterministic paired trials with seeds 2–51. For every trial, the same generated truth, measurements, communication realization, and delivered measurement sets are used for all compared fusion arms within that trial.

B. Additional Attempted Modules and Results

This appendix records secondary routes and modules that were evaluated but not retained as part of the main claimed method. The purpose is to document that the design space was explored beyond the main GA/KLA method family and to bound the empirical value of alternatives that are not promoted into the paper body.

B.1. AA-based secondary route

The paper body focuses on GA/KLA fusion because the method is designed around conservative logarithmic pooling and branch-specific GA-LMB fusion weights. A secondary arithmetic-average experiment was run with the earlier stabilized weight configuration. It is retained only as design-history evidence and is not used to support the current no-stabilization Balanced configuration. The diagnostic shortens the AA run horizon and applies stronger AA-specific pruning to keep mixture growth tractable.

Table 5

Appendix-only AA-based secondary route.

Arm	OSPA	Loc. disag.	Cardinality
fixed AA	3.903	5.130	0.226
covariance-link AA	3.507	4.401	0.103
stabilized branch-decoupled AA	3.449	4.306	0.102
stabilized FID-FIA AA	3.457	4.328	0.095

The stabilized adaptive AA variants reduce disagreement relative to fixed AA, but their configuration no longer matches the retained GA method. These values therefore motivate a future no-stabilization AA rerun rather than a transfer claim in the present paper.

B.2. NIS-based consistency weighting

This appendix-level experiment studies whether innovation-based consistency information should enter the adaptive weight model. NIS is a standard filter-diagnostic statistic in tracking and navigation, and recent work also uses related distributional checks for estimator auto-tuning [39, 40]. The relevant comparison here is w/o NIS $\rightarrow$ robust NIS $\rightarrow$ plain NIS. The NIS term is treated as a consistency penalty rather than a monotonic quality reward because innovation consistency is structurally coupled with covariance-based posterior quality.

Table 6

Network disagreement metrics for the NIS-based consistency comparison.

Arm	OSPA	Loc. disag.	Cardinality
w/o NIS	1.792	1.556	0.198
robust NIS	1.804	1.552	0.203
plain NIS	1.897	1.739	0.237

Plain NIS is clearly harmful in the present setting. Robust NIS remains substantially better than plain NIS, but across 50 trials it is still slightly worse than the no-NIS baseline in OSPA and cardinality dispersion, with only a small localization-disagreement improvement.

B.3. Freshness weighting

This experiment tests whether a recency-oriented score improves fusion beyond the robust-NIS baseline. The intended comparison is robust NIS baseline versus robust NIS plus freshness. In the 50-trial comparison, the result is effectively flat: OSPA remains at 1.804, matched localization disagreement changes from 1.552 to 1.551, and cardinality dispersion remains at 0.203.

B.4. History weighting

This experiment evaluates whether a temporal-stability score improves fusion quality. In the exploratory comparison, OSPA changes from 1.811 to 1.814, localization disagreement changes from 3.173 to 3.158, and cardinality dispersion changes from 0.214 to 0.215. This effect is too small and too mixed to justify a central role in the main text.

B.5. Cardinality-consensus weighting

An exploratory direct cardinality-consensus score was also tested. Relative to the covariance-and-link baseline, the pilot changed OSPA from 1.909508 to 1.954855, RMSE from 1.621662 to 1.791287, and cardinality dispersion from 0.242500 to 0.285000. Because this evidence is negative and only a pilot, it is retained only as appendix context.

B.6. Association ambiguity and posterior-structure consistency

Two additional extensions were also evaluated. The association-ambiguity add-on changes the baseline OSPA, RMSE, and cardinality dispersion from (1.874840, 1.779820, 0.244500) to (1.876368, 1.769102, 0.245500), which is too mixed to justify promotion into the main text. The stronger posterior-structure-consistency extension remains tied with the retained weak static structure prior at (1.862244, 1.749608, 0.244250), so it is best treated as a future extension candidate rather than a present claim.

Taken together, these appendix results delimit the explored design space. Only components supported by the controlled no-stabilization GA experiment are promoted into the retained method.

C. Theoretical Interpretation of Adaptive Branch-Decoupled Fusion

This appendix records the theoretical interpretation of the adaptive branch-decoupled fusion rule used in the main paper. The purpose is narrower than a full new optimality theory. We focus on statements that are supported by the proposed method and the standard KLA/LMB background [5–7, 12–15]. Throughout this appendix, we assume the same label alignment and moment-compatible Gaussian projection used by the merger.

C.1. Entropy-Regularized Interpretation of the Weight Backbone

Fix a fusion node s at time k with active neighborhood $\mathcal{A}_{k,s}$. The adaptive backbone constructs the positive score

$$s_{0,j} = q_{\text{cov},k}^{(j)} q_{\text{link},k}^{(j)} q_{\text{exist},k}^{(j)}, \quad j \in \mathcal{A}_{k,s}, \quad (29)$$

where q_{cov} , q_{link} , and q_{exist} denote the covariance-quality, realized link-quality, and existence-confidence terms, respectively. Define

$$u_{0,j} = \log q_{\text{cov},k}^{(j)} + \log q_{\text{link},k}^{(j)} + \log q_{\text{exist},k}^{(j)}. \quad (30)$$

Let

$$\Delta_{k,s} = \left\{ w \in \mathbb{R}^{|\mathcal{A}_{k,s}|} : w_j \geq 0, \sum_j w_j = 1 \right\} \quad (31)$$

be the active-neighbor simplex.

Proposition C1. For any temperature $\tau > 0$, the optimization problem

$$\max_{w \in \Delta_{k,s}} \left\{ \sum_j w_j u_{0,j} + \tau H(w) \right\}, \quad H(w) = - \sum_j w_j \log w_j, \quad (32)$$

has the softmax solution

$$w_j^\star = \frac{\exp(u_{0,j}/\tau)}{\sum_u \exp(u_{0,u}/\tau)} = \frac{s_{0,j}^{1/\tau}}{\sum_u s_{0,u}^{1/\tau}}. \quad (33)$$

In particular, when $\tau = 1$, the implemented normalized product rule is recovered:

$$w_j^\star = \frac{s_{0,j}}{\sum_u s_{0,u}}. \quad (34)$$

Thus the multiplicative backbone can be read as an entropy-regularized simplex allocation in log-utility space, not merely as an empirical product of factors.

Isotropic interpretation of the covariance term. If the local Gaussian components satisfy

$$T_{k,i}^{(j)} = \sigma_j^2 I_d \quad (35)$$

for all labels i at sensor j , then the implemented covariance score

$$q_{\text{cov},k}^{(j)} = \frac{1}{\epsilon + d\sigma_j^2} \quad (36)$$

induces the same ordering as smaller σ_j^2 , smaller $\log \det T_{k,i}^{(j)}$, larger $\log \det (T_{k,i}^{(j)})^{-1}$, and larger $\text{tr}((T_{k,i}^{(j)})^{-1})$. Hence, in the isotropic limit, the retained trace-based score agrees with standard information criteria. In near-isotropic regimes, the disagreement with log-determinant criteria enters only through higher-order anisotropy terms.

C.2. Algebraic Branch Structure of the Implemented GA-LMB Merger

The implemented merger uses separate spatial and existence weights. This separation is not purely heuristic; it is supported by the algebraic structure of the GA-LMB update.

Spatial branch. For a fixed Bernoulli component i , the Gaussian canonical fusion step is

$$K_i = \sum_j w_{x,j} T_{i,j}^{-1}, \quad (37)$$

$$h_i = \sum_j w_{x,j} T_{i,j}^{-1} v_{i,j}, \quad (38)$$

with

$$\Sigma_i = K_i^{-1}, \quad \mu_i = \Sigma_i h_i. \quad (39)$$

Proposition C2. Conditional on the spatial weights w_x , the Gaussian density $\mathcal{N}(\mu_i, \Sigma_i)$ is obtained as the minimizer of

$$\min_{q \in \mathcal{G}} \sum_j w_{x,j} D_{\text{KL}}(q \parallel \mathcal{N}(v_{i,j}, T_{i,j})), \quad (40)$$

where $\mathcal{G}$ is the Gaussian family. Hence, under the stated projection assumption, the implemented spatial update is the Gaussian KLA barycenter.

Existence branch. The same merger computes

$$r_i = \frac{\eta_i(w_x) \prod_j r_{i,j}^{w_{r,j}}}{\eta_i(w_x) \prod_j r_{i,j}^{w_{r,j}} + \prod_j (1 - r_{i,j})^{w_{r,j}}}, \quad (41)$$

where $\eta_i(w_x)$ is the spatial-overlap normalization term induced by the Gaussian branch.

Proposition C3. The fused existence probability satisfies the log-odds identity

$$\log \frac{r_i}{1 - r_i} = \log \eta_i(w_x) + \sum_j w_{r,j} \log \frac{r_{i,j}}{1 - r_{i,j}}. \quad (42)$$

Therefore, for fixed spatial weights, the existence branch is a weighted logit pool with an additive spatial-overlap offset.

C.3. From Bernoulli-RFS Decomposition to an Aligned-LMB Surrogate

The key theoretical bridge is that Bernoulli-RFS divergence already separates into existence and spatial terms.

Proposition C4. Let $\beta = (r, p)$ and $\beta_j = (r_j, p_j)$ be Bernoulli RFS densities. Then

$$D_{\text{KL}}(\beta \parallel \beta_j) = d_{\text{Ber}}(r \parallel r_j) + r D_{\text{KL}}(p \parallel p_j), \quad (43)$$

where

$$d_{\text{Ber}}(r \parallel r_j) = (1 - r) \log \frac{1 - r}{1 - r_j} + r \log \frac{r}{r_j}. \quad (44)$$

This decomposition shows that spatial mismatch is naturally weighted by existence level.

Proposition C5. For a fixed Bernoulli component i , consider the decoupled surrogate

$$\mathcal{J}_i(r, p; w_x, w_r) = \sum_j w_{r,j} d_{\text{Ber}}(r \parallel r_{i,j}) + r \sum_j w_{x,j} D_{\text{KL}}(p \parallel p_{i,j}). \quad (45)$$

Its minimizer satisfies

$$p_i^\star(x) = \frac{1}{\eta_i(w_x)} \prod_j p_{i,j}(x)^{w_{x,j}}, \quad (46)$$

$$\log \frac{r_i^\star}{1 - r_i^\star} = \log \eta_i(w_x) + \sum_j w_{r,j} \log \frac{r_{i,j}}{1 - r_{i,j}}. \quad (47)$$

Hence, under the stated aligned-label and projection assumptions, the implemented spatial and existence update rules solve a per-Bernoulli decoupled surrogate objective.

If local posteriors share an aligned label set and are represented in product LMB form, then the reverse-KL divergence is additive across labels. Under this same alignment assumption, the per-Bernoulli surrogate lifts directly to an aligned-LMB surrogate objective that decomposes labelwise.

Scope boundary. This appendix does not claim that the branch-decoupled rule is the unique optimizer of a fully unconstrained LMB fusion objective. The exact derivation above still relies on the same two architectural assumptions used by the merger: aligned labels and moment-compatible spatial projection.

C.4. Branch-Decoupled Weight Allocation and Structure-Aware Regularization

The branch-specific scores are

$$s_{x,j} = (q_{\text{cov},j})^{\alpha_x} (q_{\text{link},j})^{\beta_x}, \quad (48)$$

$$s_{r,j} = (q_{\text{link},j})^{\beta_r} (q_{\text{exist},j})^{\alpha_r}, \quad (49)$$

with geometric interpolation

$$\tilde{s}_{x,j} = s_{0,j}^{1-\eta_x} s_{x,j}^{\eta_x}, \quad (50)$$

$$\tilde{s}_{r,j} = s_{0,j}^{1-\eta_r} s_{r,j}^{\eta_r}. \quad (51)$$

This is a convex interpolation in log-utility space. Define $\hat{u}_{x,j} = \log \tilde{s}_{x,j}$ and $\hat{u}_{r,j} = \log \tilde{s}_{r,j}$ for the branch utilities before graph and FID-FIA refinement.

The FID-FIA existence refinement adds one more nonnegative factor only to the existence branch. Let a_j denote the Fisher-information-distance accumulation score computed from existence-weighted local target pairs, and define the bounded normalized score

$$q_{\text{fid},j} = \rho_{\min} + (1 - \rho_{\min}) \left(\frac{a_j}{\epsilon + \max_{u \in \mathcal{A}_{k,s}} a_u} \right)^{p_f}. \quad (52)$$

The refined existence score is

$$\tilde{s}_{r,j}^{\text{fid}} = \tilde{s}_{r,j} q_{\text{fid},j}^{\gamma_f}. \quad (53)$$

In log-utility form, this is simply

$$\hat{u}_{r,j}^{\text{fid}} = \hat{u}_{r,j} + \gamma_f \log q_{\text{fid},j}, \quad (54)$$

while the spatial utility $\hat{u}_{x,j}$ is unchanged. The log-utility interpretation below is direct for $\rho_{\min} > 0$; the main experiment setting $\rho_{\min} = 0$ is the boundary limit that allows the existence branch to place zero instantaneous weight on sensors whose FID-FIA accumulation is zero.

After decoupling, the structure-aware refinement multiplies the spatial score by $\xi_{x,j}^{\gamma_x}$ and the existence score, or the FID-FIA-refined existence score when enabled, by $\xi_{r,j}^{\gamma_r}$. This is equivalent to adding $\gamma_x \log \xi_{x,j}$ and $\gamma_r \log \xi_{r,j}$ to the corresponding branch utilities.

Proposition C6. Let p_x and p_r be normalized priors satisfying

$$p_{x,j} \propto \xi_{x,j}^{\gamma_x/\tau_x}, \quad (55)$$

$$p_{r,j} \propto \xi_{r,j}^{\gamma_r/\tau_r}. \quad (56)$$

Then the branch-wise optimization problems are equivalently written as

$$w_x^* = \arg \max_{w \in \Delta_{k,s}} \left\{ \sum_j w_j \hat{u}_{x,j} - \tau_x D_{\text{KL}}(w \| p_x) \right\}, \quad (57)$$

$$w_r^* = \arg \max_{w \in \Delta_{k,s}} \left\{ \sum_j w_j \hat{u}_{r,j} - \tau_r D_{\text{KL}}(w \| p_r) \right\}. \quad (58)$$

Thus the weak structure-aware term can be interpreted as a KL regularization toward a graph-induced prior on the simplex.

Corollary C6a. With the FID-FIA existence refinement, the same branch-wise entropy-regularized program remains valid after replacing the existence utility $\hat{u}_{r,j}$ by $\hat{u}_{r,j}^{\text{fid}}$. Equivalently,

$$w_{r,j}^* \propto \tilde{s}_{r,j} \xi_{r,j}^{\gamma_r} q_{\text{fid},j}^{\gamma_f}, \quad w_{x,j}^* \propto \tilde{s}_{x,j} \xi_{x,j}^{\gamma_x}. \quad (59)$$

Thus the FID-FIA term is theoretically just another existence-branch log-utility, not a change to the spatial KLA barycenter.

Proposition C7. The statement below applies to the unrefined existence utility $\hat{u}_{r,j}$ and, after the substitution in Corollary C6a, to the FID-FIA-refined utility $\hat{u}_{r,j}^{\text{fid}}$. Define the branch utilities

$$U_x(w_x) = \sum_j w_{x,j} \hat{u}_{x,j} + \tau_x H(w_x), \quad (60)$$

$$U_r(w_r) = \sum_j w_{r,j} \hat{u}_{r,j} + \tau_r H(w_r). \quad (61)$$

Then, under the stated aligned-label and projection assumptions, the branch-decoupled adaptive weights solve the block-separable surrogate program

$$\max_{w_x, w_r \in \Delta_{k,s}} \{U_x(w_x) + U_r(w_r)\}. \quad (62)$$

Moreover, together with the aligned-LMB surrogate state-update block above, the merger and the adaptive weight-allocation rule are consistent with a single block-relaxed objective whose two blocks reproduce the state-update and weight-update steps.

Scalar-FID-FIA as a constrained special case. Using a single FID-FIA scalar weight vector for the whole fused posterior can be viewed as imposing the additional constraint $w_x = w_r = w_{\text{fid}}$ and choosing both branch utilities to be dominated by $\log q_{\text{fid},j}$. The proposed refinement relaxes this constraint: w_x remains governed by the communication-aware spatial utility, while w_r receives the FID-FIA term. This provides a theoretical explanation for the observed empirical tradeoff. A single FID-FIA weight can improve cardinality by moving the existence logit pool toward sensors with high pairwise Fisher separability, but the same move also perturbs the Gaussian precision barycenter. Branch decoupling keeps the information-geometric signal where it is most useful, in the Bernoulli existence pool, without forcing the spatial barycenter to use the same weights.

FID-FIA as an existence-branch separability utility. The aligned-LMB surrogate in Proposition C5 suggests that spatial and existence weights should respond to different failure modes: spatial fusion is tied to existence-weighted spatial mismatch, while existence fusion is tied to Bernoulli existence mismatch. The retained covariance, realized-link, and existence-confidence utilities are stable local proxies for these sensitivities. The FID-FIA accumulation is used in the narrower role of an information-geometric separability cue for Bernoulli existence decisions: it favors sensors whose local target-pair geometry is more distinguishable under the Fisher metric induced by the sensor model. It should therefore be read as an existence-branch utility, not as an exact Bernoulli KL gradient or as a full objective for unconstrained LMB fusion.

C.5. Existence Confidence

The existence-confidence term uses

$$c(r) = |2r - 1|, \quad (63)$$

which is bounded in $[0, 1]$. This quantity is a bounded monotone surrogate of Bernoulli decisiveness because

$$\left| \log \frac{r}{1-r} \right| = \log \frac{1+c(r)}{1-c(r)} = 2 \operatorname{artanh}(c(r)). \quad (64)$$

Therefore the existence-confidence score is a bounded monotone proxy for the magnitude of the Bernoulli natural parameter that appears in the existence branch (42).

CRedit authorship contribution statement

Hao Lang: Conceptualization, Supervision, Project administration, Writing - review and editing. **Jinhao Chen:** Software, Validation, Formal analysis, Writing - original draft. **Tianyu Wo:** Supervision, Writing - review and editing.

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Data availability

No real-world datasets were used in this study. The simulation data supporting the findings are synthetic and can be regenerated from the MATLAB/Octave scripts and deterministic seeds used in the experiments. The code and generated result logs will be made available by the corresponding author on reasonable request.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT by OpenAI to improve the clarity, grammar, and readability of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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